Truncated higher Stark elements and p-adic class-number divisibility for almost abelian extensions of imaginary quadratic fields

Abstract

Let k be an imaginary quadratic field and let F/k be finite abelian, with Galois group G. The torsion-free unit lattice $E_F = \mathcal{O}_F^\times/\mu(F)$ and the augmentation kernel $X_G \subset \mathbb{Z}[G]$ become isomorphic over $\mathbb{R}$ through the Dirichlet embedding, while the product of the first derivatives $L'(0,\chi)$ of the non-trivial Artin L-functions of F/k is a rational multiple of the Dirichlet regulator R_F. The Stark–Dirichlet module formed by these two lattices in the determinant line has generalized index

$$\frac{h_F}{h_k} \cdot \mathfrak{z}(F/k),$$

where $\mathfrak{z}(F/k)$ is the explicit annihilator index of $\mu(F)$ against $\mu(k)$. For primes p not dividing $|G|$, idempotents of $\mathbb{Z}_p[G]$ cut the same integer out of the non-trivial isotypic components of the class group. Ramification-complete truncation of the L-functions is necessary for an integral higher Stark element to meet the equivariant leading value: if a ramified place is omitted and the non-decomposition hypothesis fails there, the χ-rank of the smaller S-unit lattice is strictly less than the order of vanishing, so the top exterior power in the leading degree is zero. Under the non-decomposition hypothesis the truncation multiplier is the integer $\prod_v |G/I_v|$, it is a p-unit for every $p \nmid |G|$, and the p-part of h_F/h_k equals the p-part of the Stark–Dirichlet index corrected by roots of unity.

1 Introduction

Equivariant refinements of Stark's conjecture ask for an element of an exterior power of S-units whose regulator recovers the leading term, at $s = 0$, of an Artin L-function, together with an integrality statement in the group ring. Tate's formulation [10] isolates the leading coefficient as a lattice index in a Minkowski space built from places. Burns–Flach [1] recast the same comparison as an equivariant Tamagawa number, and Ritter–Weiss [5] pursued the integral group-ring statement through Iwasawa theory. Rubin's higher Stark elements [8] address vanishing of order greater than one by passing to exterior powers. Over an imaginary quadratic base, finite-order Hecke L-functions have a single archimedean gamma factor of type $\Gamma(s)$, so the analytic side of this comparison is unusually rigid: every non-trivial Artin L-function of an abelian extension F/k has a simple zero at $s = 0$, and the product of the derivatives is pinned down by the analytic class number formula.

The present note stays inside that rigid setting and develops a self-contained lattice comparison. The base field k is imaginary quadratic, the extension F/k is finite abelian, and $F/\mathbb{Q}$ is not required to be Galois. The last point is the reason for the name adopted here: an *almost abelian datum* is a pair (F/k) of this type. Complex conjugation on $k/\mathbb{Q}$ need not extend to F, so every equivariant construction below is a module for $G = \mathrm{Gal}(F/k)$ alone.

Three statements are proved. The first identifies a generalized index built from L-derivatives and the Dirichlet regulator with a ratio of class numbers times an annihilator index of roots of unity.

The second decomposes the p-adic story, for $p \nmid |G|$, by idempotents, and shows that an integral higher Stark element representing the ramification-complete leading value cannot live at a truncation level that omits a ramified place at which Frobenius acts trivially on some non-trivial inertia-fixed character. The third converts the non-decomposition case into an explicit integer multiplier and a p-adic divisibility for h_F/h_k.

Recent work on integral equivariant leading-term conjectures has emphasized the denominators created when ramified Euler factors are left in place. Those investigations are the motivation for isolating the imaginary-quadratic abelian case. The arguments below do not rely on them. The only external inputs are the analytic class number formula at $s = 0$, as in Neukirch [3, Chapter VII], and Hecke's functional equation for finite-order L-functions of k [2, 3]. Everything about truncation, idempotents, and class-group invariants is proved directly. Individual ratios $L'(0, \chi)/R(\chi)$ are not claimed to lie in the character field. Only the product over all non-trivial characters, divided by the full regulator, is shown to be rational. Where a step needs $p \nmid |G|$ or a non-decomposition hypothesis, the hypothesis is stated and used.

1.1 Statement of the main theorems

Write $n = |G|$, $w_K = |\mu(K)|$, and $h_K = |\operatorname{Cl}_K|$. Let $E_F = \mathcal{O}_F^\times/\mu(F)$, a free $\mathbb{Z}$-module of rank $n - 1$, and let

$$X_G = \ker\big(\mathbb{Z}[G] \xrightarrow{\varepsilon} \mathbb{Z}\big)$$

be the augmentation kernel. Section 3 constructs a G-equivariant Dirichlet embedding $\lambda_F : E_F \to X_G \otimes \mathbb{R}$ whose covolume on the determinant line is the Dirichlet regulator R_F. The product of leading coefficients

$$\mathcal{L}(F/k) = \prod_{\chi \in \widehat{G} \setminus \{\mathbf{1}\}} L'(0, \chi)$$

is then compared with that covolume. The positive rational

$$\mathfrak{z}(F/k) = \big[\operatorname{Ann}_{\mathbb{Z}}(\mu(k)) : \operatorname{Ann}_{\mathbb{Z}}(\mu(F))\big]^{-1} = \frac{w_k}{w_F}$$

is the roots-of-unity annihilator index.

Theorem 1.1 (Stark–Dirichlet index)**.** *Let (F/k) be an almost abelian datum. The Stark–Dirichlet module $\mathrm{SD}(F/k)$ of Definition 4.1 is a pair of rank-one lattices in the determinant line of $X_G \otimes \mathbb{R}$, and its generalized index against the torsion-free unit lattice satisfies*

$$\operatorname{Ind}\big(\mathrm{SD}(F/k),\, E_F\big) = \frac{\mathcal{L}(F/k)}{R_F} = \frac{h_F}{h_k} \cdot \mathfrak{z}(F/k) \in \mathbb{Q}_{>0}.$$

A finite set S of places of k is a *truncation set* when it contains the unique archimedean place ∞. It is *ramification-complete* when it also contains every finite place ramified in F/k. The truncated L-function $L_S(s, \chi)$ is obtained by removing the Euler factors at the finite places in S at which χ is unramified. For a ramified place v, write $I_v \subset D_v \subset G$ for the inertia and decomposition groups of a place of F above v; both are well-defined subgroups because G is abelian. The *non-decomposition hypothesis* ND_v asserts that G/I_v is cyclic and generated by the Frobenius element of v.

Theorem 1.2 (p-adic higher Stark lattices)**.** *Let p be a prime not dividing $|G|$, and let ψ run over the primitive idempotents of $\mathbb{Q}_p[G]$.*

(i) *The group ring decomposes as a product of unramified discrete valuation rings, $\mathbb{Z}_p[G] \cong \prod_\psi \mathcal{O}_\psi$, with idempotents $e_\psi \in \mathbb{Z}_p[G]$.*

(ii) *For every truncation set $S \ni \infty$ and every non-trivial ψ, the module $e_\psi(E_{F,S} \otimes \mathbb{Z}_p)$ is free of rank*

$$d_\psi(S) = \operatorname{ord}_{s=0} L_S(s, \chi)$$

over $\mathcal{O}_\psi$, where χ is any complex character in the orbit cut out by e_ψ. The higher Stark module $\mathrm{HSt}^{(p)}_{S,\psi}(F/k) = \det_{\mathcal{O}_\psi}\big(e_\psi(E_{F,S} \otimes \mathbb{Z}_p)\big)$ is therefore free of rank one over $\mathcal{O}_\psi$.

(iii) *Let v be ramified in F/k, let $S_\star$ be a ramification-complete truncation set, and let $S = S_\star \setminus \{v\}$. If ND_v fails, then for some non-trivial ψ one has $d_\psi(S) < d_\psi(S_\star)$. Every element of $\mathrm{HSt}^{(p)}_{S,\psi}(F/k)$ has vanishing image in the determinant line of degree $d_\psi(S_\star)$, while the leading coefficient $L^*_{S_\star}(0, \chi)$ is nonzero. No integral higher Stark element of level S maps to the ψ-component of the ramification-complete equivariant leading value.*

In particular, ramification-complete truncation is necessary for the existence of such an integral element whenever some omitted ramified place fails ND_v.

Theorem 1.3 (Obstruction and p-adic divisibility)**.** *Let (F/k) be an almost abelian datum and let S_{ram} be the set of finite places of k ramified in F/k.*

(i) (Order obstruction.) *If $v \in S_{\mathrm{ram}}$ is omitted from a truncation set $S \ni \infty$ and ND_v fails, then there exists $\chi \in \widehat{G} \setminus \{\mathbf{1}\}$ with*

$$\operatorname{ord}_{s=0} L_{S\cup\{v\}}(s, \chi) > \dim_{\mathbb{C}}\big(E_{F,S} \otimes \mathbb{C}\big)_\chi.$$

No integral higher Stark element of level S maps to the ramification-complete leading value.

(ii) (Truncation multiplier.) *Assume ND_v for every $v \in S_{\mathrm{ram}}$. Then each local factor $1 - \chi(\varphi_v)$ is nonzero, the integer*

$$\mathfrak{c}(F/k) = \prod_{v \in S_{\mathrm{ram}}} |G/I_v|$$

equals $\prod_{\chi \neq \mathbf{1}} L'_{S_{\mathrm{ram}}}(0, \chi)/\mathcal{L}(F/k)$, and the ramification-truncated index satisfies

$$\mathrm{Ind}^{\mathrm{ram}}(F/k) := \frac{1}{R_F} \prod_{\chi \neq \mathbf{1}} L'_{S_{\mathrm{ram}}}(0, \chi) = \mathfrak{c}(F/k) \cdot \frac{h_F}{h_k} \cdot \mathfrak{z}(F/k) \in \mathbb{Q}_{>0}.$$

(iii) (p-adic divisibility.) *Assume ND_v for every $v \in S_{\mathrm{ram}}$, and let $p \nmid |G|$. Then $\mathfrak{c}(F/k)$ is a p-adic unit, and*

$$\begin{aligned} v_p\Big(\frac{h_F}{h_k}\Big) &= \sum_{\psi \neq \mathbf{1}} v_p\big(\big|e_\psi(\mathrm{Cl}_F \otimes \mathbb{Z}_p)\big|\big) \\ &= v_p\big(\mathrm{Ind}(\mathrm{SD}(F/k), E_F)\big) + v_p(w_F) - v_p(w_k) \\ &= v_p\big(\mathrm{Ind}^{\mathrm{ram}}(F/k)\big) + v_p(w_F) - v_p(w_k) \geq 0. \end{aligned}$$

Thus the p-primary part of h_k divides the p-primary part of h_F.

The inequality $v_p(h_F) \geq v_p(h_k)$ uses $p \nmid |G|$ in an essential way. Section 7 records the Hilbert class field of $\mathbb{Q}(\sqrt{-5})$, where $|G| = 2$ and $v_2(h_F) < v_2(h_k)$. The same section checks the identity of Theorem 1.1 on three extensions by comparing class-number data with derivatives of Dedekind zeta functions.

1.2 Organization

Section 2 fixes notation. Section 3 develops the Dirichlet embedding, the Euler-product form of Artin formalism, simple zeros at $s = 0$, and the character formula for orders of truncated L-functions. Section 4 defines the Stark–Dirichlet module and proves Theorem 1.1. Section 5 treats idempotents and higher Stark lattices and proves Theorem 1.2. Section 6 proves the obstruction and the divisibility. Section 7 records the numerical checks, and Section 8 lists open questions.

2 Notation and almost abelian data

Definition 2.1. An *almost abelian datum* is a finite abelian extension F/k in which the base field k is imaginary quadratic. Write $G = \mathrm{Gal}(F/k)$ and $n = |G|$. The extension $F/\mathbb{Q}$ is not required to be Galois.

Throughout, places of k and of F are denoted v and w respectively. The field k has a unique archimedean place ∞, which is complex. A finite place v of k is *ramified* in F/k when the inertia group of any place of F above v is nontrivial. Let $S_{\mathrm{ram}} = S_{\mathrm{ram}}(F/k)$ denote the finite set of such places. For $v \in S_{\mathrm{ram}}$ the inertia group I_v and the decomposition group D_v are subgroups of G. Abelianness forces every subgroup to be normal, so these groups do not depend on the choice of place above v, and the Frobenius element φ_v is a well-defined element of D_v/I_v.

Definition 2.2. The place v satisfies the *non-decomposition hypothesis* ND_v when the quotient G/I_v is cyclic and generated by φ_v. Equivalently, $D_v/I_v = G/I_v$, so v does not decompose in the fixed field of I_v.

Since D_v/I_v is always cyclic, generated by φ_v, the hypothesis ND_v fails if and only if either G/I_v is non-cyclic or φ_v does not generate it. In that case some non-trivial character of G/I_v kills φ_v.

For a number field K write $\mu(K)$ for its group of roots of unity, $w_K = |\mu(K)|$, $h_K = |\operatorname{Cl}_K|$, and R_K for the Dirichlet regulator. The unit rank of k is zero; the convention $R_k = 1$ is used, so that the class number formula at $s = 0$ reads $\zeta_k(0) = -h_k/w_k$. Set

$$E_K = \mathcal{O}_K^{\times}/\mu(K).$$

Dirichlet's unit theorem gives $E_F \cong \mathbb{Z}^{n-1}$. Indeed $r_1(F) = 0$ and $r_2(F) = [F : \mathbb{Q}]/2 = n$, so the unit rank is $r_1 + r_2 - 1 = n - 1$.

Characters of G are homomorphisms $\chi : G \to \mathbb{C}^{\times}$, and $\widehat{G}$ denotes the full character group. The trivial character is $\mathbf{1}$. The idempotent attached to χ in $\mathbb{C}[G]$ is

$$e_\chi = \frac{1}{n} \sum_{g \in G} \chi(g^{-1})\, g.$$

For a $\mathbb{C}[G]$-module V write $V_\chi = e_\chi V$.

S-units are denoted $\mathcal{O}_{F,S}^{\times}$, and $E_{F,S} = \mathcal{O}_{F,S}^{\times}/\mu(F)$. If S is a finite set of places of k containing ∞, and $S(F)$ is the set of places of F above S, the S-unit theorem gives

$$\mathrm{rank}_{\mathbb{Z}}\, E_{F,S} = |S(F)| - 1.$$

The annihilator and Fitting ideal of a finite $\mathbb{Z}$-module M are written $\mathrm{Ann}_{\mathbb{Z}}(M)$ and $\mathrm{Fitt}_{\mathbb{Z}}(M)$. For $M \cong \mathbb{Z}/m\mathbb{Z}$ one has $\mathrm{Ann}_{\mathbb{Z}}(M) = \mathrm{Fitt}_{\mathbb{Z}}(M) = (m)$. Generalized indices of lattices are positive real numbers, defined in Section 4; the main theorems prove that the indices under consideration are rational.

3 Equivariant L-values, regulators, and truncation

3.1 Archimedean places and the Dirichlet–Artin embedding

Lemma 3.1. *The group G acts simply transitively on the archimedean places of F. The decomposition group of every archimedean place is trivial.*

Proof. The completion of k at ∞ is isomorphic to $\mathbb{C}$. For each place w of F above ∞ the local extension F_w/k_∞ is an extension of $\mathbb{C}$, hence has degree 1, and the decomposition group has order 1. The formula $\sum_{w|\infty}[F_w : k_\infty] = [F : k] = n$ therefore yields exactly n archimedean places. The action of G on these places is transitive, the stabilizers are the decomposition groups, and $|G| = n$, so the action is simply transitive. □

Fix an archimedean place w_0 of F and an embedding $\iota_0 : F \hookrightarrow \mathbb{C}$ inducing w_0 and extending a chosen embedding of k. Lemma 3.1 identifies the set of archimedean places with G by $g \mapsto gw_0$. The normalized absolute value at gw_0 is

$$\|u\|_{gw_0} = \big|\iota_0(g^{-1}u)\big|^2,$$

so that $\log \|u\|_{gw_0} = 2\log|\iota_0(g^{-1}u)|$. For $u \in F^\times$ define

$$\lambda_F(u) = \sum_{g\in G} \log \|u\|_{gw_0} \cdot g \in \mathbb{R}[G]. \tag{1}$$

Lemma 3.2. *The map λ_F sends $\mathcal{O}_F^\times$ into $X_G \otimes \mathbb{R}$, intertwines the left G-actions, and induces an injection $E_F \hookrightarrow X_G \otimes \mathbb{R}$. Extended $\mathbb{R}$-linearly, it is an isomorphism of $\mathbb{R}[G]$-modules*

$$\lambda_F : E_F \otimes \mathbb{R} \xrightarrow{\sim} X_G \otimes \mathbb{R}.$$

Consequently, for every $\chi \in \widehat{G}$,

$$\dim_{\mathbb{C}}(E_F \otimes \mathbb{C})_\chi = \begin{cases} 0 & \text{if } \chi = \mathbf{1}, \\ 1 & \text{if } \chi \neq \mathbf{1}. \end{cases}$$

Proof. The product formula for the field norm gives $\prod_{g\in G} \|u\|_{gw_0} = |N_{F/\mathbb{Q}}(u)|$. If u is a unit this product equals 1, so the sum of the logarithms vanishes and $\lambda_F(u)$ lies in $X_G \otimes \mathbb{R}$.

For equivariance, take $h \in G$. Left multiplication on $\mathbb{R}[G]$ gives

$$h \cdot \lambda_F(u) = \sum_{g\in G} \log \|u\|_{h^{-1}gw_0} \cdot g.$$

The embedding attached to $h^{-1}gw_0$ is $\iota_0 \circ (h^{-1}g)^{-1} = \iota_0 \circ g^{-1}h$, and since G is abelian one has $g^{-1}h = hg^{-1}$. Thus

$$\log \|u\|_{h^{-1}gw_0} = \log\big|\iota_0(g^{-1}hu)\big|^2 = \log \|hu\|_{gw_0},$$

which is the coefficient of g in $\lambda_F(hu)$. Hence $\lambda_F(hu) = h \cdot \lambda_F(u)$.

If $\lambda_F(u) = 0$, then every archimedean absolute value of u equals 1. The powers of u are algebraic integers in F of bounded house, so they form a finite set, and $u^a = u^b$ for some $a > b$. Thus u is a root of unity. The induced map on E_F is therefore injective. Both $E_F \otimes \mathbb{R}$ and $X_G \otimes \mathbb{R}$ have dimension $n - 1$, so the $\mathbb{R}$-linear extension is an isomorphism. It is G-equivariant by the calculation above.

Finally, $\mathbb{C}[G] = \bigoplus_\chi \mathbb{C}e_\chi$ and the augmentation $\varepsilon \otimes \mathbb{C}$ is the projection onto the trivial line, so $X_G \otimes \mathbb{C} = \bigoplus_{\chi\neq\mathbf{1}} \mathbb{C}e_\chi$. The isomorphism λ_F transports these multiplicities to E_F. □

Lemma 3.3. *Let δ_E be a $\mathbb{Z}$-basis vector of $\det_{\mathbb{Z}} E_F = \bigwedge_{\mathbb{Z}}^{n-1} E_F$ and let δ_X be a $\mathbb{Z}$-basis vector of $\det_{\mathbb{Z}} X_G$. Normalize the Haar measure on the hyperplane $X_G \otimes \mathbb{R}$ so that X_G has covolume 1 with respect to δ_X. Then*

$$\det(\lambda_F)(\delta_E) = \pm R_F \, \delta_X,$$

where R_F is the Dirichlet regulator of F. In particular the covolume of $\lambda_F(E_F)$ equals R_F.

Proof. Index archimedean places by group elements and choose the ordered $\mathbb{Z}$-basis $\{g - 1 : g \neq 1\}$ of X_G, up to sign. If $\varepsilon_1, \dots, \varepsilon_{n-1}$ is a $\mathbb{Z}$-basis of a complement of $\mu(F)$ in $\mathcal{O}_F^\times$, the coordinates of $\lambda_F(\varepsilon_j)$ are

$$c_{g,j} = 2 \log \bigl| \iota_0(g^{-1} \varepsilon_j) \bigr|, \qquad g \in G,$$

and $\sum_g c_{g,j} = 0$. Deleting the coordinate at the identity, the resulting $(n-1) \times (n-1)$ matrix is the classical regulator matrix of a totally complex field: one real coordinate $2 \log |\sigma(\varepsilon)|$ for each of $n-1$ archimedean places. The absolute value of its determinant is R_F by the standard normalization of the regulator [3, Chapter I]. The relation $\sum_g c_{g,j} = 0$ implies that every maximal minor has the same absolute value, and that this absolute value is the covolume of $\lambda_F(E_F)$ in the hyperplane for the measure giving X_G covolume 1. Hence $\det(\lambda_F)(\delta_E) = \pm R_F \, \delta_X$. □

3.2 Artin L-functions through the Euler product

For $\chi \in \widehat{G}$ the Artin L-function of the one-dimensional representation χ is the Euler product

$$L(s, \chi) = \prod_{\substack{v \\ \chi|_{I_v} = \mathbf{1}}} \bigl(1 - \chi(\varphi_v) \, Nv^{-s}\bigr)^{-1}, \tag{2}$$

absolutely convergent for $\mathrm{Re}(s) > 1$. At a ramified place with $\chi|_{I_v} \neq \mathbf{1}$ the local factor is 1 and is omitted from the product. Class field theory identifies χ with a primitive finite-order Hecke character of k whose conductor is supported exactly on the places where χ is ramified, and (2) is the Hecke L-function of that character. In particular $L(s, \mathbf{1}) = \zeta_k(s)$.

Lemma 3.4. *As meromorphic functions on $\mathbb{C}$,*

$$\zeta_F(s) = \prod_{\chi \in \widehat{G}} L(s, \chi).$$

Proof. It is enough to compare Euler factors. Let v be a finite place of k, set $T = Nv^{-s}$, and write f for the order of φ_v in $\Gamma = G/I_v$ and $g_\Gamma = |\Gamma|/f$. The places of F above v all have residue degree f over v after the inertia extension is taken into account, and the Euler factor of ζ_F at v is $(1 - T^f)^{-|\Gamma|/f}$. More precisely, the inertia subgroup contributes the ramification index $|I_v|$ uniformly, the decomposition group satisfies $|D_v/I_v| = f$, and the number of places above v is $|G|/|D_v| = |\Gamma|/f$, each with absolute norm Nv^f. Thus the Euler factor is $(1 - T^f)^{-|\Gamma|/f}$.

On the Artin side only characters trivial on I_v contribute a non-trivial local factor, and those characters are the characters of Γ. Left multiplication by φ_v on the group algebra $\mathbb{C}[\Gamma]$ decomposes into $|\Gamma|/f$ cycles of length f, so its characteristic polynomial is $(X^f - 1)^{|\Gamma|/f}$. The regular representation of an abelian group is the sum of its characters, hence

$$\prod_{\psi \in \widehat{\Gamma}} \bigl(Y - \psi(\varphi_v)\bigr) = (Y^f - 1)^{|\Gamma|/f}.$$

Substituting $Y = T^{-1}$ and clearing powers of T yields

$$\prod_{\psi \in \widehat{\Gamma}} \bigl(1 - \psi(\varphi_v)\, T\bigr) = (1 - T^f)^{|\Gamma|/f}.$$

Inverting both sides matches the Euler factor of ζ_F. Characters that are non-trivial on I_v contribute the empty local factor 1 on the Artin side and do not appear in $\widehat{\Gamma}$. The identity of Euler products for $\mathrm{Re}(s) > 1$ extends by analytic continuation. □

Definition 3.5. Let S be a truncation set. For $\chi \in \widehat{G}$ the *S-truncated L-function* is

$$L_S(s, \chi) = L(s, \chi) \prod_{\substack{v \in S \setminus \{\infty\} \\ \chi|_{I_v} = \mathbf{1}}} \bigl(1 - \chi(\varphi_v)\, Nv^{-s}\bigr). \tag{3}$$

The product is finite. If χ is ramified at v, that place contributes no factor.

Removing the same Euler factors from every character in Lemma 3.4 gives the companion identity

$$\zeta_{F,S}(s) = \prod_{\chi \in \widehat{G}} L_S(s, \chi), \tag{4}$$

where $\zeta_{F,S}$ is the Dedekind zeta function with the Euler factors at places above $S \setminus \{\infty\}$ removed.

3.3 Simple zeros

The input from Hecke is the following form of the functional equation. A constant factor 2, sometimes inserted so that the gamma factor reads $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s)$, does not affect zeros or the non-vanishing argument below and is omitted.

Hypothesis 3.6 (Hecke)**.** Let ψ be a non-trivial primitive finite-order Hecke character of k, of conductor $\mathfrak{f}$. The completed function

$$\Lambda(s, \psi) = \bigl(|D_k|\, N\mathfrak{f}\bigr)^{s/2} (2\pi)^{-s} \Gamma(s)\, L(s, \psi)$$

is entire, and $\Lambda(s, \psi) = W(\psi)\, \Lambda(1 - s, \overline{\psi})$ for a constant $W(\psi)$ of absolute value 1.

This is Hecke's theorem [2]; a modern account is given in [3, Chapter VII]. The next lemma is the structural fact that makes the imaginary-quadratic case rigid.

Lemma 3.7. *Let $\chi \in \widehat{G} \setminus \{\mathbf{1}\}$. Then $L(s, \chi)$ is entire,* $\mathrm{ord}_{s=0} L(s, \chi) = 1$*, and $L'(0, \chi) \neq 0$. Moreover $L(1, \chi) \neq 0$.*

Proof. By the Euler product (2) and the identification recalled above, $L(s, \chi)$ is the primitive Hecke L-function of a non-trivial finite-order character of k. Hypothesis 3.6 supplies entirety.

Lemma 3.4 gives $\prod_{\eta \neq \mathbf{1}} L(s, \eta) = \zeta_F(s)/\zeta_k(s)$. Both Dedekind zeta functions have a single simple pole at $s = 1$, with nonzero residues, so the quotient extends holomorphically through $s = 1$ and takes a nonzero value there. Each factor $L(s, \eta)$ for $\eta \neq \mathbf{1}$ is entire, hence holomorphic at $s = 1$. A product of holomorphic functions that is nonzero at a point has every factor nonzero at that point. Thus $L(1, \chi) \neq 0$, and the same applies to $\overline{\chi}$.

Hypothesis 3.6 yields $\Lambda(0, \chi) = W(\chi)\, \Lambda(1, \overline{\chi})$. The value

$$\Lambda(1, \overline{\chi}) = \bigl(|D_k|\, N\mathfrak{f}\bigr)^{1/2} (2\pi)^{-1} \Gamma(1)\, L(1, \overline{\chi})$$

is nonzero because $\Gamma(1) = 1$ and $L(1, \overline{\chi}) \neq 0$. Hence $\Lambda(0, \chi) \neq 0$. The remaining factors $(|D_k|\, N\mathfrak{f})^{s/2}(2\pi)^{-s}$ are holomorphic and nonzero at $s = 0$, while $\Gamma(s)$ has a simple pole there. A simple pole in the gamma factor, cancelled in the product Λ but not over-cancelled, forces $L(s, \chi)$ to have a simple zero at $s = 0$. $\square$

Corollary 3.8. *One has*

$$\prod_{\chi \neq \mathbf{1}} L'(0, \chi) = \frac{\zeta_F^*(0)}{\zeta_k(0)} = \frac{h_F}{h_k} \cdot \frac{w_k}{w_F} \cdot R_F, \tag{5}$$

where $\zeta_F^*(0) = \lim_{s \to 0} s^{-(n-1)} \zeta_F(s)$ *and* $\zeta_k(0) = -h_k / w_k$.

Proof. The analytic class number formula [3, Chapter VII] gives

$$\zeta_F^*(0) = -\frac{h_F R_F}{w_F}, \qquad \zeta_k(0) = -\frac{h_k}{w_k},$$

the second because the unit rank of k is zero and $R_k = 1$. Both quantities are nonzero. Lemma 3.4 and Lemma 3.7 expand the product near $s = 0$:

$$\zeta_F(s) = \zeta_k(s) \prod_{\chi \neq \mathbf{1}} L(s, \chi) = \big(\zeta_k(0) + O(s)\big)\left(\left(\prod_{\chi \neq \mathbf{1}} L'(0, \chi)\right) s^{n-1} + O(s^n)\right).$$

The coefficient of s^{n-1} is $\zeta_k(0) \prod_{\chi \neq \mathbf{1}} L'(0, \chi)$. The order of ζ_F at $s = 0$ is exactly $n - 1$, so this coefficient equals $\zeta_F^*(0)$. Dividing by $\zeta_k(0)$ produces the first equality of (5), and the class number formula produces the second. The right-hand side is positive, so the product of the derivatives is positive. $\square$

No claim is made about the individual complex numbers $L'(0, \chi)$. Their product is rational after division by R_F, because the right-hand side of (5) is rational times R_F.

3.4 Orders after truncation

Lemma 3.9. *Let S be a truncation set and let $\chi \in \widehat{G} \setminus \{\mathbf{1}\}$. Set*

$$Z(S, \chi) = \big\{ v \in S \setminus \{\infty\} : \chi|_{I_v} = \mathbf{1} \text{ and } \chi(\varphi_v) = 1 \big\}.$$

Then

$$\operatorname{ord}_{s=0} L_S(s, \chi) = 1 + |Z(S, \chi)|,$$

and the leading coefficient is

$$L_S^*(0, \chi) = L'(0, \chi) \left(\prod_{\substack{v \in S \setminus \{\infty\} \\ \chi|_{I_v} = \mathbf{1} \\ \chi(\varphi_v) \neq 1}} \big(1 - \chi(\varphi_v)\big) \right) \left(\prod_{v \in Z(S, \chi)} \log Nv \right). \tag{6}$$

In particular, if $Z(S, \chi)$ is empty, then $\operatorname{ord}_{s=0} L_S(s, \chi) = 1$ *and*

$$L_S'(0, \chi) = L'(0, \chi) \prod_{\substack{v \in S \setminus \{\infty\} \\ \chi|_{I_v} = \mathbf{1}}} \big(1 - \chi(\varphi_v)\big).$$

Proof. Start from (3) and the simple zero of Lemma 3.7. For each finite place in the product, the expansion

$$1 - \chi(\varphi_v)\, Nv^{-s} = 1 - \chi(\varphi_v) + \chi(\varphi_v)\, s \log Nv + O(s^2)$$

is holomorphic. If $\chi(\varphi_v) \neq 1$ the value at $s = 0$ equals $1 - \chi(\varphi_v) \neq 0$, so the order is unchanged and the leading coefficient is multiplied by that constant. If $\chi(\varphi_v) = 1$ the factor equals $s \log Nv + O(s^2)$, a simple zero with leading coefficient $\log Nv$. Places at which χ is ramified do not appear. Adding these contributions to the simple zero of $L(s, \chi)$ gives the order formula and (6). □

The same count governs S-units.

Lemma 3.10. *Let S be a truncation set and let $\chi \in \widehat{G} \setminus \{\mathbf{1}\}$. Then*

$$\dim_{\mathbb{C}}\big(E_{F,S} \otimes \mathbb{C}\big)_{\chi} = \#\{v \in S : \chi|_{D_v} = \mathbf{1}\} = 1 + |Z(S, \chi)| = \operatorname{ord}_{s=0} L_S(s, \chi).$$

The trivial character has multiplicity $|S| - 1$ in $E_{F,S} \otimes \mathbb{C}$.

Proof. The G-equivariant Dirichlet embedding on S-units identifies $E_{F,S} \otimes \mathbb{R}$ with the augmentation kernel $X_{S(F)} \otimes \mathbb{R}$ inside $\mathbb{R}[S(F)]$, by the S-unit theorem and the same equivariance argument as in Lemma 3.2. The permutation module $\mathbb{Z}[S(F)]$ decomposes as $\bigoplus_{v \in S} \operatorname{Ind}_{D_v}^{G} \mathbb{Z}$. For a one-dimensional character the Frobenius reciprocity computation

$$\langle \operatorname{Ind}_{D_v}^{G} \mathbf{1},\, \chi \rangle = \langle \mathbf{1},\, \chi|_{D_v} \rangle$$

equals 1 if $\chi|_{D_v} = \mathbf{1}$ and equals 0 otherwise. The augmentation $X_{S(F)} \to 0$ is the kernel of a map onto the trivial representation, so it does not change multiplicities of non-trivial characters. Therefore the multiplicity of χ is the number of $v \in S$ with $\chi|_{D_v} = \mathbf{1}$.

The archimedean decomposition group is trivial by Lemma 3.1, so χ always contributes the place ∞. For a finite place, $\chi|_{D_v} = \mathbf{1}$ if and only if $\chi|_{I_v} = \mathbf{1}$ and $\chi(\varphi_v) = 1$, which is the condition defining $Z(S, \chi)$. Lemma 3.9 identifies the resulting integer with the order of vanishing. The multiplicity of the trivial character in $\mathbb{Z}[S(F)]$ equals $|S|$, and passage to the augmentation kernel reduces it by one. □

Corollary 3.11. *For every truncation set S and every $\chi \in \widehat{G}$, the order of vanishing of $L_S(s, \chi)$ at $s = 0$ equals the multiplicity of χ in $E_{F,S} \otimes \mathbb{C}$, except that both sides are interpreted for $\chi = \mathbf{1}$ as $|S| - 1$ on the unit side and as the order of $\zeta_{k,S}$ on the L-function side. In particular the equality for non-trivial characters holds whether or not S contains S_{ram}.*

The corollary is deliberately weaker than a recommendation to omit ramified places. Rank matching at a small truncation set S controls L_S, not the leading term of a larger truncation. Section 5 uses the gap between the two ranks when a ramified place is restored.

3.5 The roots-of-unity annihilator

Since $\mu(k) \subset \mu(F)$ and both groups are cyclic, w_k divides w_F.

Lemma 3.12. *The quotient $\mu(F)/\mu(k)$ is cyclic of order w_F/w_k, and*

$$\operatorname{Ann}_{\mathbb{Z}}\big(\mu(F)/\mu(k)\big) = \Big(\frac{w_F}{w_k}\Big).$$

Equivalently,

$$\mathfrak{z}(F/k) := \big[\operatorname{Ann}_{\mathbb{Z}}(\mu(k)) : \operatorname{Ann}_{\mathbb{Z}}(\mu(F))\big]^{-1} = \frac{w_k}{w_F} = \big[\mathbb{Z} : \operatorname{Ann}_{\mathbb{Z}}(\mu(F)/\mu(k))\big]^{-1}.$$

One also has $\mathrm{Fitt}_{\mathbb{Z}}(\mu(F)) = (w_F)$ *and* $\mathrm{Fitt}_{\mathbb{Z}}(\mu(k)) = (w_k)$*, so* $\mathfrak{z}(F/k)$ *is the ratio of the positive generators of these Fitting ideals.*

Proof. A finite subgroup of $\mathbb{C}^\times$ is cyclic, so $\mu(F)$ is cyclic of order w_F and contains a unique subgroup of each dividing order. That subgroup of order w_k is $\mu(k)$. The quotient of cyclic groups is cyclic of order w_F/w_k, and the annihilator of $\mathbb{Z}/m\mathbb{Z}$ is (m). The identification $\mathrm{Ann}_{\mathbb{Z}}(\mu(K)) = (w_K)$ gives

$$\big[\mathrm{Ann}_{\mathbb{Z}}(\mu(k)) : \mathrm{Ann}_{\mathbb{Z}}(\mu(F))\big] = \big[w_k\mathbb{Z} : w_F\mathbb{Z}\big] = \frac{w_F}{w_k},$$

and inverting this index produces $\mathfrak{z}(F/k)$. The Fitting ideals of cyclic groups coincide with their annihilators. □

The group G acts on $\mu(F)$. The invariant subgroup is $\mu(F)^G = \mu(F) \cap k = \mu(k)$, so $H^0(G, \mu(F))$ has order w_k. The scalar $\mathfrak{z}(F/k)$ is exactly the ratio of this invariant order to $|\mu(F)|$. A finer group-ring Fitting ideal of $\mu(F)$ would see the cyclotomic character of the action; the index in Theorem 1.1 uses only the underlying abelian group, which is what the class number formula produces.

4 The Stark–Dirichlet module

Definition 4.1. Let δ_X be a generator of $\det_{\mathbb{Z}} X_G \cong \mathbb{Z}$, and equip $X_G \otimes \mathbb{R}$ with the Haar measure of Lemma 3.3. The *regulator lattice* and the *leading lattice* in the real line $\mathbb{R}\delta_X$ are

$$\Lambda_{\mathrm{reg}}(F/k) = R_F\,\mathbb{Z} \cdot \delta_X,$$
$$\Lambda_{\mathrm{lead}}(F/k) = \mathcal{L}(F/k)\,\mathbb{Z} \cdot \delta_X,$$

where $\mathcal{L}(F/k) = \prod_{\chi \neq \mathbf{1}} L'(0, \chi)$. The *Stark–Dirichlet module* $\mathrm{SD}(F/k)$ is the ordered pair of these two lattices. Its *generalized index* against the torsion-free unit lattice is the ratio of positive generators

$$\mathrm{Ind}\big(\mathrm{SD}(F/k),\, E_F\big) = \frac{\mathcal{L}(F/k)}{R_F} \in \mathbb{R}_{>0}. \tag{7}$$

The generator δ_X cancels in the ratio, so the index does not depend on that choice. Lemma 3.3 is the statement that the covolume of $\lambda_F(E_F)$ is the generator of Λ_{reg}.

Proof of Theorem 1.1. Corollary 3.8 gives

$$\frac{\mathcal{L}(F/k)}{R_F} = \frac{h_F}{h_k} \cdot \frac{w_k}{w_F}.$$

Lemma 3.12 identifies w_k/w_F with $\mathfrak{z}(F/k)$. The right-hand side lies in $\mathbb{Q}_{>0}$, and the left-hand side is the generalized index (7). □

The proof is short because the preparatory lemmas carry the analytic content. What the theorem asserts is the equality of two independently defined quantities: the lattice index formed from L-derivatives and $\lambda_F(E_F)$, and the arithmetic ratio $(h_F/h_k)\mathfrak{z}(F/k)$.

Corollary 4.2. *The generalized index* $\mathrm{Ind}(\mathrm{SD}(F/k), E_F)$ *lies in* $\mathbb{Q}_{>0}$*. For every prime* p*,*

$$v_p\big(\mathrm{Ind}(\mathrm{SD}(F/k), E_F)\big) = v_p(h_F) - v_p(h_k) + v_p(w_k) - v_p(w_F). \tag{8}$$

Remark 4.3. The index need not lie in $\mathbb{Z}$. For the Hilbert class field of $\mathbb{Q}(\sqrt{-5})$ the value is $1/4$; see Section 7. Integrality of a Stark element that hits $\mathcal{L}(F/k)$ on the nose is a stronger requirement, measured by whether this rational number lies in $\mathbb{Z}$, and is not claimed.

For later comparison, record the reduced-norm form of the same identity. The equivariant leading element in the complex group ring is

$$\Theta(F/k) = \sum_{\chi \neq \mathbf{1}} L'(0, \chi^{-1})\, e_\chi \in \mathbb{C}[G]. \tag{9}$$

Its reduced norm on the reduced regular representation $\bigoplus_{\chi \neq \mathbf{1}} \mathbb{C}$ is

$$\mathrm{Nrd}\big(\Theta(F/k)\big) = \prod_{\chi \neq \mathbf{1}} L'(0, \chi^{-1}) = \mathcal{L}(F/k),$$

the second equality by reindexing $\chi \mapsto \chi^{-1}$. Theorem 1.1 is therefore also the statement

$$\frac{\mathrm{Nrd}\big(\Theta(F/k)\big)}{R_F} = \frac{h_F}{h_k} \cdot \text{ы}(F/k).$$

5 Idempotents and higher Stark lattices

5.1 The group ring at primes away from the group order

Lemma 5.1. *Let p be a prime not dividing $n = |G|$. Then $\mathbb{Z}_p[G]$ is a finite étale $\mathbb{Z}_p$-algebra. It decomposes as a product of discrete valuation rings*

$$\mathbb{Z}_p[G] \cong \prod_{\psi} \mathcal{O}_\psi,$$

indexed by the primitive idempotents e_ψ of $\mathbb{Q}_p[G]$, where each $\mathcal{O}_\psi$ is the valuation ring of a finite unramified extension of $\mathbb{Q}_p$. The idempotents e_ψ lie in $\mathbb{Z}_p[G]$.

Proof. A finite abelian group is a direct product of cyclic groups, and $\mathbb{Z}[G \times H] \cong \mathbb{Z}[G] \otimes_{\mathbb{Z}} \mathbb{Z}[H]$. It is therefore enough to treat G cyclic of order n and then observe that a tensor product of finite étale $\mathbb{Z}_p$-algebras is finite étale.

For G cyclic, $\mathbb{Z}_p[G] \cong \mathbb{Z}_p[X]/(X^n - 1)$. In $\mathbb{F}_p[X]$ the derivative of $X^n - 1$ is nX^{n-1}. If a root of $X^n - 1$ in an algebraic closure of $\mathbb{F}_p$ were a common root of the derivative, then p would divide n, which is excluded. Thus $X^n - 1$ is separable over $\mathbb{F}_p$. Hensel's lemma lifts its factorization into distinct irreducibles to a factorization of $X^n - 1$ in $\mathbb{Z}_p[X]$ as a product of monic polynomials with unit discriminant. The Chinese remainder theorem yields

$$\mathbb{Z}_p[X]/(X^n - 1) \cong \prod_i \mathbb{Z}_p[X]/(f_i),$$

and each $\mathbb{Z}_p[X]/(f_i)$ is the valuation ring of the unramified extension of $\mathbb{Q}_p$ cut out by f_i. An unramified extension has a unique prime above p, so each factor is a discrete valuation ring. The corresponding idempotents are the images of $(1, 0, \ldots)$, $(0, 1, \ldots)$, and so on, and therefore lie in $\mathbb{Z}_p[G]$. □

Each idempotent e_ψ is a sum of complex idempotents e_χ over a $\mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$-orbit of characters, after any embedding of the cyclotomic values into $\overline{\mathbb{Q}}_p$. Because $p \nmid n$, the conductor of each character divides a power of an integer coprime to p, so these cyclotomic extensions are unramified at p, in agreement with Lemma 5.1.

Lemma 5.2. *Fix $p \nmid n$ and a truncation set S. The function $\chi \mapsto \operatorname{ord}_{s=0} L_S(s,\chi)$ is constant on $\mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$-orbits of characters. If e_ψ cuts out such an orbit Ω_ψ of non-trivial characters, then for every $\chi \in \Omega_\psi$*

$$\dim_{K_\psi} e_\psi(E_{F,S} \otimes \mathbb{Q}_p) = \operatorname{ord}_{s=0} L_S(s,\chi),$$

where $K_\psi = \mathrm{Frac}(\mathcal{O}_\psi)$.

Proof. An automorphism σ fixing $\mathbb{Q}_p$ acts on character values by $\sigma(\chi)(g) = \sigma(\chi(g))$. The equality $\sigma(\chi)(\varphi_v) = 1$ holds if and only if $\sigma(\chi(\varphi_v)) = 1$, hence if and only if $\chi(\varphi_v) = 1$, because σ fixes 1. The same applies to the restriction of χ to I_v. Lemma 3.9 therefore gives the same order of vanishing at every character of the orbit.

The complex multiplicity of each such χ in $E_{F,S} \otimes \mathbb{C}$ equals that order, by Lemma 3.10. The $\mathbb{Q}_p$-dimension of $e_\psi(E_{F,S} \otimes \mathbb{Q}_p)$ equals $|\Omega_\psi|$ times this common multiplicity, while $[K_\psi : \mathbb{Q}_p] = |\Omega_\psi|$. Dividing yields the rank over K_ψ. □

5.2 Higher Stark modules

Definition 5.3. Let $p \nmid |G|$, let S be a truncation set, and let e_ψ be a non-trivial primitive idempotent of $\mathbb{Z}_p[G]$. The module $e_\psi(E_{F,S} \otimes \mathbb{Z}_p)$ is finitely generated and torsion-free over the discrete valuation ring $\mathcal{O}_\psi$. It is therefore free, of some rank $d_\psi(S)$. The *higher Stark module* of level S at ψ is the determinant

$$\mathrm{HSt}^{(p)}_{S,\psi}(F/k) = \bigwedge_{\mathcal{O}_\psi}^{d_\psi(S)} e_\psi\big(E_{F,S} \otimes \mathbb{Z}_p\big),$$

a free $\mathcal{O}_\psi$-module of rank one. An *integral higher Stark element* of level S at ψ is an $\mathcal{O}_\psi$-basis vector of this module.

Freeness over a discrete valuation ring is the reason for restricting to $p \nmid |G|$. At primes dividing $|G|$ the group ring is not étale, torsion-free modules need not be free, and the determinant line requires a derived construction that this note does not undertake.

Proof of Theorem 1.2. Assertion (i) is Lemma 5.1.

For (ii), Lemma 5.2 computes the generic rank $d_\psi(S)$ as $\operatorname{ord}_{s=0} L_S(s,\chi)$. A finitely generated torsion-free module over a discrete valuation ring is free of rank equal to the dimension of its generic fiber. The top exterior power of a free module of that rank is free of rank one. This is $\mathrm{HSt}^{(p)}_{S,\psi}(F/k)$.

For (iii), let v be ramified, let $S_\star$ contain v, and set $S = S_\star \setminus \{v\}$. Assume ND_v fails. By Definition 2.2 there exists a non-trivial character χ of G/I_v with $\chi(\varphi_v) = 1$. Viewed as a character of G, one has $\chi|_{I_v} = \mathbf{1}$, $\chi \neq \mathbf{1}$, and $\chi(\varphi_v) = 1$, so $v \in Z(S_\star, \chi)$ while $v \notin Z(S,\chi)$. Lemma 3.9 gives

$$\operatorname{ord}_{s=0} L_{S_\star}(s,\chi) = \operatorname{ord}_{s=0} L_S(s,\chi) + 1.$$

Let ψ be the idempotent whose orbit contains χ. Every Galois conjugate of χ likewise satisfies $\sigma(\chi)(\varphi_v) = 1$ and is trivial on I_v, so the common order of vanishing on the orbit increases by exactly 1. Lemma 5.2 identifies d_ψ with that common order, and therefore $d_\psi(S_\star) = d_\psi(S) + 1$.

The higher Stark module of level S is an exterior power of degree $d_\psi(S)$. The ramification-complete leading coefficient $L^*_{S_\star}(0,\chi)$ is the nonzero generator of a determinant line of degree $d_\psi(S_\star) > d_\psi(S)$, in the sense of the covolume on $\bigwedge^{d_\psi(S_\star)}(E_{F,S_\star} \otimes \mathbb{R})_\chi$. The exterior power of degree $d_\psi(S_\star)$ of a free module of rank $d_\psi(S)$ is zero. Equivalently, if $V \subset W$ are vector spaces over

the fraction field of $\mathcal{O}_\psi$ with $\dim V < \dim W$, then $\bigwedge^{\dim W} V = 0$, so every element of the level-S determinant has image zero in the level-$S_\star$ determinant line. The leading coefficient is nonzero by definition of the order of vanishing. Hence no integral higher Stark element of level S maps to that leading value.

Ramification-complete truncation is exactly the operation of restoring every such v to the truncation set, which raises the rank of $\mathrm{HSt}^{(p)}_{S,\psi}$ to the degree in which the leading coefficient lives. Without that restoration the integral element does not exist. □

Remark 5.4. If instead ND_v holds, Lemma 3.9 shows that restoring v does not increase $\mathrm{ord}_{s=0} L_S(s,\chi)$ for any $\chi \neq \mathbf{1}$. The non-trivial higher Stark modules of level S and of level $S \cup \{v\}$ have the same rank. Truncation is then invisible at the level of ranks and is felt only through the algebraic multiplier of Theorem 1.3(ii). The necessity statement in Theorem 1.2(iii) is the complementary case, in which the multiplier is replaced by a jump in order and the integral element in the smaller degree is impossible. Both cases are required for a complete description; neither is optional in the statement.

5.3 What the integral element is asked to match

It remains to spell out the meaning of "maps to the equivariant leading value", used in Theorem 1.2(iii) and Theorem 1.3(i).

Definition 5.5. Let $S_\star$ be ramification-complete and let $\chi \neq \mathbf{1}$ satisfy $m = \mathrm{ord}_{s=0} L_{S_\star}(s,\chi)$. A class η in a top exterior power of a χ-isotypic unit lattice *maps to* the leading value $L^*_{S_\star}(0,\chi)$ when the following two conditions hold.

(a) The χ-rank d of the lattice equals m, so that the regulator embedding identifies the determinant line of the lattice with the determinant line of $(X_{S_\star(F)} \otimes \mathbb{R})_\chi$.

(b) Under that identification, the image of η equals $L^*_{S_\star}(0,\chi)$ times the fixed volume form coming from the Haar measure that gives the augmentation lattice covolume 1.

If $d < m$, condition (a) fails and η does not map to the leading value.

Condition (b) is the usual Stark regulator identity in a single character component, assembled equivariantly by taking one integral element in each higher Stark module $\mathrm{HSt}^{(p)}_{S,\psi}$. Theorem 1.2 does not assert that an element satisfying (b) exists. Existence of a generator of the free module $\mathrm{HSt}^{(p)}_{S_\star,\psi}$ is unconditional; equality of its regulator with the leading coefficient is a form of the equivariant Stark conjecture and is not proved here. What is proved is the obstruction: when the degree is wrong, condition (a) already fails, so no choice of integral element can satisfy the identity.

6 Obstruction and class-number divisibility

6.1 The order obstruction

Proof of Theorem 1.3(i). The argument is the character form of Theorem 1.2(iii), without p-adic coefficients. Since ND_v fails, there exists $\chi \neq \mathbf{1}$ with $v \in Z(S \cup \{v\}, \chi)$ and $v \notin Z(S,\chi)$. Lemmas 3.9 and 3.10 give

$$\mathrm{ord}_{s=0} L_{S\cup\{v\}}(s,\chi) = 1 + |Z(S\cup\{v\},\chi)| > 1 + |Z(S,\chi)| = \dim_{\mathbb{C}}(E_{F,S} \otimes \mathbb{C})_\chi.$$

The top exterior power of the χ-component of $E_{F,S}$ in this larger degree vanishes, while $L^*_{S\cup\{v\}}(0,\chi) \neq 0$. Definition 5.5 therefore excludes every integral element of level S. □

The same conclusion can be phrased without choosing χ. If ND_v fails, the permutation module on places of F above v contributes a non-trivial character to $X_{(S\cup\{v\})(F)}$, and the S-unit lattice at level S maps into the proper subspace of the larger Minkowski space on which the new place has coordinate zero. A covolume in the larger degree cannot be represented by a wedge of vectors from the smaller subspace.

6.2 The truncation multiplier

Lemma 6.1. *Let Γ be a finite abelian group and let $\varphi \in \Gamma$. Then*

$$\prod_{\psi\in\widehat{\Gamma}\setminus\{\mathbf{1}\}} \big(1-\psi(\varphi)\big) = \begin{cases} |\Gamma| & \text{if } \Gamma = \langle\varphi\rangle, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. The product vanishes if and only if some non-trivial character kills φ, which happens if and only if φ lies in a maximal subgroup, if and only if $\langle\varphi\rangle \neq \Gamma$. If $\Gamma = \langle\varphi\rangle$, then Γ is cyclic of some order f and the values $\psi(\varphi)$ run through all f-th roots of unity exactly once. The polynomial identity $X^f - 1 = \prod_{j=0}^{f-1}(X - \zeta^j)$ gives, after dividing by $X - 1$ and setting $X = 1$,

$$\prod_{j=1}^{f-1}(1-\zeta^j) = f.$$

The left-hand side is the product in the statement, and $f = |\Gamma|$. □

Proof of Theorem 1.3(ii). Assume ND_v for every ramified v. For each such v and each $\chi \neq \mathbf{1}$ with $\chi|_{I_v} = \mathbf{1}$, the character χ descends to a non-trivial character of $G/I_v = \langle\varphi_v\rangle$, so $\chi(\varphi_v) \neq 1$. Lemma 3.9 therefore keeps the order of $L_{S_{\mathrm{ram}}}(s,\chi)$ equal to 1, and

$$L'_{S_{\mathrm{ram}}}(0,\chi) = L'(0,\chi) \prod_{\substack{v\in S_{\mathrm{ram}} \\ \chi|_{I_v}=\mathbf{1}}} \big(1-\chi(\varphi_v)\big). \tag{10}$$

Characters ramified at v contribute an empty factor at that place. Taking the product over $\chi \neq \mathbf{1}$ and rearranging gives

$$\prod_{\chi\neq\mathbf{1}} \frac{L'_{S_{\mathrm{ram}}}(0,\chi)}{L'(0,\chi)} = \prod_{v\in S_{\mathrm{ram}}} \prod_{\substack{\chi\neq\mathbf{1} \\ \chi|_{I_v}=\mathbf{1}}} \big(1-\chi(\varphi_v)\big).$$

For fixed v the inner product is the product of Lemma 6.1 for $\Gamma = G/I_v$ and $\varphi = \varphi_v$. The hypothesis ND_v is exactly $\Gamma = \langle\varphi\rangle$, so the inner product equals $|G/I_v|$. Hence

$$\prod_{\chi\neq\mathbf{1}} L'_{S_{\mathrm{ram}}}(0,\chi) = \mathfrak{c}(F/k)\,\mathcal{L}(F/k),$$

with $\mathfrak{c}(F/k) = \prod_{v\in S_{\mathrm{ram}}} |G/I_v|$. Each $|G/I_v|$ is a positive integer, and an empty product equals 1, so $\mathfrak{c}(F/k) \in \mathbb{Z}_{>0}$. Dividing by R_F and applying Theorem 1.1 yields

$$\mathrm{Ind}^{\mathrm{ram}}(F/k) = \mathfrak{c}(F/k) \cdot \frac{h_F}{h_k} \cdot \mathfrak{z}(F/k).$$

The right-hand side is rational and positive. □

Corollary 6.2. *Assume* ND_v *for every* $v \in S_{\mathrm{ram}}$*, and assume* $\mathfrak{c}(F/k) \neq 1$*. Let* η_0 *be a* $\mathbb{Z}$*-basis vector of* $\det_{\mathbb{Z}} E_F$*, so that the regulator of* η_0 *equals* R_F *in absolute value. Then* η_0 *cannot map to both* $\mathcal{L}(F/k)$ *and* $\prod_{\chi \neq \mathbf{1}} L'_{S_{\mathrm{ram}}}(0, \chi)$*. An integral element that maps to one of these two leading products differs from an integral element that maps to the other by the integer factor* $\mathfrak{c}(F/k)$ *in the determinant line.*

Proof. The two leading products differ by the integer $\mathfrak{c}(F/k) \neq 1$, by the identity in the proof of Theorem 1.3(ii). A single real number R_F cannot equal both products unless the rational index compensates, and the compensation for one product is $\mathfrak{c}(F/k)$ times the compensation for the other. □

Corollary 6.2 is the non-decomposition form of the obstruction. Omitting the ramified places replaces the target leading product by its quotient by $\mathfrak{c}(F/k)$. An element integral at a prime divisor of $\mathfrak{c}(F/k)$ and primitive in the determinant line can hit at most one of the two targets. Combined with Theorem 1.3(i), one obtains a complete dichotomy. If ND_v fails, the integral element of the smaller level does not exist in the right degree. If ND_v holds, the element may exist in the right degree, but the omitted-truncation leading value and the ramification-complete leading value differ by the integer $|G/I_v|$, so integrality relative to a primitive generator cannot serve both targets when that integer is greater than 1.

6.3 Class-group invariants away from the group order

Lemma 6.3. *Let* $p \nmid |G|$*. Extension of ideals defines a homomorphism* $j : \mathrm{Cl}_k \to \mathrm{Cl}_F^G$*, and the field norm defines* $N : \mathrm{Cl}_F \to \mathrm{Cl}_k$*. After tensoring with* $\mathbb{Z}_p$ *these maps become inverse isomorphisms up to the unit* $n = |G|$*:*

$$j : \mathrm{Cl}_k \otimes \mathbb{Z}_p \xrightarrow{\sim} (\mathrm{Cl}_F \otimes \mathbb{Z}_p)^G,$$

with inverse $n^{-1}N$*. In particular,*

$$v_p\big(\big|(\mathrm{Cl}_F \otimes \mathbb{Z}_p)^G\big|\big) = v_p(h_k)$$

and

$$v_p(h_F) - v_p(h_k) = v_p\big(\big|e'(\mathrm{Cl}_F \otimes \mathbb{Z}_p)\big|\big) \geq 0, \tag{11}$$

where $e' = 1 - e_{\mathbf{1}}$ *is the sum of the non-trivial idempotents.*

Proof. If $\mathfrak{b}$ is an ideal of $\mathcal{O}_k$, the extended ideal $\mathfrak{b}\mathcal{O}_F$ is stable under G, so j lands in the invariant classes. If $\mathfrak{a}$ is an ideal of $\mathcal{O}_F$, the product of its conjugates equals $(N_{F/k}\mathfrak{a})\,\mathcal{O}_F$, hence $j \circ N$ is multiplication by the norm element $N_G = \sum_{g \in G} g$ on Cl_F. In the other direction, $N_{F/k}(\mathfrak{b}\mathcal{O}_F) = \mathfrak{b}^n$, so $N \circ j$ is multiplication by n on Cl_k.

Since $p \nmid n$, the scalar n is a unit in $\mathbb{Z}_p$, and the trivial idempotent $e_{\mathbf{1}} = N_G/n$ lies in $\mathbb{Z}_p[G]$ by Lemma 5.1. For every $\mathbb{Z}_p[G]$-module M the invariant submodule equals the image of this idempotent: $M^G = e_{\mathbf{1}}M$. Indeed an invariant vector is fixed by $e_{\mathbf{1}}$, and a vector in the image of $e_{\mathbf{1}}$ is fixed by G because $e_{\mathbf{1}}$ is central and $ge_{\mathbf{1}} = e_{\mathbf{1}}$. Taking G-invariants is therefore an exact functor on $\mathbb{Z}_p[G]$-modules, realized by multiplication with $e_{\mathbf{1}}$.

On $\mathrm{Cl}_k \otimes \mathbb{Z}_p$ the composition $n^{-1}N \circ j$ is the identity. On $(\mathrm{Cl}_F \otimes \mathbb{Z}_p)^G$ the composition $j \circ n^{-1}N$ equals $n^{-1}N_G = e_{\mathbf{1}}$, which acts as the identity on invariants. Thus j and $n^{-1}N$ are inverse isomorphisms between $\mathrm{Cl}_k \otimes \mathbb{Z}_p$ and $(\mathrm{Cl}_F \otimes \mathbb{Z}_p)^G$. Comparing orders gives $v_p(|(\mathrm{Cl}_F \otimes \mathbb{Z}_p)^G|) = v_p(h_k)$.

Lemma 5.1 supplies a decomposition of the finite module $\mathrm{Cl}_F \otimes \mathbb{Z}_p$ as the product of its idempotent components, so

$$\big|\mathrm{Cl}_F \otimes \mathbb{Z}_p\big| = \big|e_{\mathbf{1}}(\mathrm{Cl}_F \otimes \mathbb{Z}_p)\big| \cdot \big|e'(\mathrm{Cl}_F \otimes \mathbb{Z}_p)\big|.$$

The first factor on the right has p-valuation $v_p(h_k)$, and the left-hand side has p-valuation $v_p(h_F)$. Subtracting produces (11). The valuation of an order is nonnegative. □

The hypothesis $p \nmid |G|$ cannot be omitted. Section 7 gives an extension with $|G| = 2$, $h_k = 2$ and $h_F = 1$.

Proof of Theorem 1.3(iii). Assume ND_v for all ramified v, and let $p \nmid |G|$. Each integer $|G/I_v|$ divides $|G|$, so $v_p(\mathfrak{c}(F/k)) = 0$. Theorem 1.3(ii) therefore yields

$$v_p\big(\mathrm{Ind}^{\mathrm{ram}}(F/k)\big) = v_p\big(\mathrm{Ind}(\mathrm{SD}(F/k), E_F)\big).$$

Corollary 4.2 rewrites the untruncated valuation as

$$v_p\big(\mathrm{Ind}(\mathrm{SD}(F/k), E_F)\big) = v_p(h_F) - v_p(h_k) + v_p(w_k) - v_p(w_F),$$

which rearranges to

$$v_p\Big(\frac{h_F}{h_k}\Big) = v_p\big(\mathrm{Ind}(\mathrm{SD}(F/k), E_F)\big) + v_p(w_F) - v_p(w_k).$$

The same equality holds with $\mathrm{Ind}^{\mathrm{ram}}$ in place of the untruncated index.

Lemma 6.3 decomposes the class-group difference by idempotents. Since $\mathbb{Z}_p[G] \cong \prod_\psi \mathcal{O}_\psi$, one has

$$\big|e'(\mathrm{Cl}_F \otimes \mathbb{Z}_p)\big| = \prod_{\psi \neq \mathbf{1}} \big|e_\psi(\mathrm{Cl}_F \otimes \mathbb{Z}_p)\big|,$$

and taking p-valuations produces the sum in the statement. Nonnegativity is (11). In particular $v_p(h_F) \geq v_p(h_k)$, so the p-primary part of h_k divides the p-primary part of h_F. □

The three expressions in Theorem 1.3(iii) are the content of the divisibility. The sum over idempotents is a decomposition of the p-part of the relative class number. The Stark–Dirichlet index computes the same integer after the roots-of-unity correction $v_p(w_F) - v_p(w_k)$ is inserted. Non-decomposition enters so that the ramification-truncated index, which is the leading term attached to the truncation level at which higher Stark elements are formed, has the same p-valuation. If some ND_v fails, Lemma 3.9 introduces a positive power of $\log Nv$ into the leading coefficient, the truncated index ceases to be rational, and the comparison with h_F/h_k is not a p-adic valuation identity.

Corollary 6.4. *Under the hypotheses of Theorem 1.3(iii), if moreover $p \nmid w_F$, then*

$$v_p\Big(\frac{h_F}{h_k}\Big) = v_p\big(\mathrm{Ind}^{\mathrm{ram}}(F/k)\big) = \sum_{\psi \neq \mathbf{1}} v_p\big(\big|e_\psi(\mathrm{Cl}_F \otimes \mathbb{Z}_p)\big|\big).$$

Proof. If $p \nmid w_F$, then $p \nmid w_k$ as well, because w_k divides w_F. The correction terms $v_p(w_F)$ and $v_p(w_k)$ vanish. □

7 Numerical checks

The identity of Theorem 1.1 was compared, for three almost abelian data, with derivatives of Dedekind zeta functions computed in PARI/GP [4]. In each case G is cyclic of order 2, so there is one non-trivial character. Then $\mathcal{L}(F/k) = L'(0, \chi)$, and this derivative equals $\zeta_F^*(0)/\zeta_k(0)$. PARI returns Taylor derivatives; the leading coefficient of a zero of order r is the r-th derivative divided by $r!$. In all three cases the ratio of the analytic side to $(h_F/h_k)(w_k/w_F)R_F$ equals 1 through 30 decimal places. The arithmetic factor itself is an exact rational number computed from class groups and roots of unity. A script reproducing the comparison is included as `examples/verify-index.gp`.

7.1 A non-integral index, and sharpness at primes dividing the group order

Let $k = \mathbb{Q}(\sqrt{-5})$. Then $h_k = 2$ and $w_k = 2$, so $\zeta_k(0) = -1$. The Hilbert class field F is $\mathbb{Q}(\sqrt{-5}, i)$, with minimal polynomial $x^4 + 12x^2 + 16$ over $\mathbb{Q}$. This extension is unramified at every finite place, so S_{ram} is empty, $\mathfrak{c}(F/k) = 1$, and ND_v holds vacuously. The class group and roots of unity are $h_F = 1$ and $w_F = 4$. Theorem 1.1 predicts

$$\mathrm{Ind}\big(\mathrm{SD}(F/k), E_F\big) = \frac{1}{2} \cdot \frac{2}{4} = \frac{1}{4}.$$

The regulator is positive and the identity $L'(0, \chi) = R_F/4$ is the numerical content of the check. In particular the index is rational but not an integer, as allowed by Remark 4.3.

Here $|G| = 2$. For the prime $p = 2$, which divides $|G|$, one has $v_2(h_F) = 0 < 1 = v_2(h_k)$. Lemma 6.3 and Theorem 1.3(iii) both exclude this prime. The example shows that the exclusion is sharp: the inequality $v_p(h_F) \geq v_p(h_k)$ can fail when p divides $|G|$.

7.2 A ramified quadratic extension with trivial multiplier

Let $k = \mathbb{Q}(i)$, so $h_k = 1$ and $w_k = 4$, and let $F = k(\sqrt{1+i})$, with minimal polynomial $x^4 - 2x^2 + 2$ over $\mathbb{Q}$. The discriminant of F is a power of 2, so the unique ramified finite place lies above 2. The Galois group has order 2. A ramified place in a cyclic extension of prime order has inertia equal to the whole group, hence $|G/I_v| = 1$ and the product of Lemma 6.1 over non-trivial characters trivial on I_v is an empty product, equal to 1. Truncation at this place does not change $L(s, \chi)$. Both class numbers equal 1 and $w_F = 4$, so Theorem 1.1 predicts that the index equals 1, and the numerical comparison agrees.

7.3 A prime that divides the base class number but not the group order

Let $k = \mathbb{Q}(\sqrt{-23})$, so $h_k = 3$ and $w_k = 2$, and let $F = k(\sqrt{2})$, with minimal polynomial x^4+42x^2+625 over $\mathbb{Q}$. Then $|G| = 2$, the prime $p = 3$ does not divide $|G|$, and the computation gives $h_F = 6$, $w_F = 2$. Theorem 1.3(iii) predicts $v_3(h_F) \geq v_3(h_k)$. Both valuations equal 1, so the non-trivial idempotent component of $\mathrm{Cl}_F \otimes \mathbb{Z}_3$ is trivial and the divisibility holds with equality of valuations. The arithmetic factor is $(6/3) \cdot (2/2) = 2$, and the comparison of $L'(0, \chi)$ with $2R_F$ again agrees to working precision. This is a case in which the p-part of h_k divides the p-part of h_F for a prime $p \nmid |G|$, with the Stark–Dirichlet index equal to 2 carrying no further 3-power.

8 Remarks and open questions

The product formula of Theorem 1.1 is unconditional. It does not factor canonically into a product of rational numbers indexed by characters, because a proper subproduct of the derivatives $L'(0, \chi)$ need not be a rational multiple of the covolume of the corresponding isotypic component. Stark's conjecture asserts precisely that each ratio $L'(0, \chi)/R(F/k, \chi)$ is algebraic and lies in the field generated by the values of χ [9, 10]. Over imaginary quadratic fields the rank-one case is tied to elliptic units [6, 7], but the lattice index proved here does not produce those units and does not require them.

Remark 8.1 (On the scope of the divisibility). The nonnegativity $v_p(h_F) \geq v_p(h_k)$ for $p \nmid |G|$ is a statement about ideal classes, proved by comparing the norm and the extension map after n has become invertible. The Stark–Dirichlet index enters when one wants the same integer as a valuation of L-derivatives divided by the regulator, corrected by w_F/w_k. Non-decomposition is the condition

under which ramified truncation multiplies the L-derivative by a rational integer rather than by a positive power of $\log Nv$. Without it, the truncated leading coefficient is not a rational multiple of R_F, and there is nothing to compare p-adically with h_F/h_k.

Several natural extensions lie outside the proofs given here.

(1) **Primes dividing $|G|$.** Lemma 5.1 fails when p divides n, and the example of Section 7 shows that the class-number inequality can reverse. A derived determinant over the order $\mathbb{Z}_p[G]$, in the sense of the equivariant Tamagawa number conjecture [1], might restore a statement with an explicit error ideal supported at p. The note does not construct that determinant.

(2) **Algebraicity of isotypic ratios.** Determine whether $\prod_{\chi\in\Omega} L'(0,\chi)$, as Ω runs over $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-orbits, becomes rational after division by the covolume of $e_\Omega\lambda_F(E_F)$ and by the discriminant of the character field. Theorem 1.1 is the case of a single orbit consisting of all non-trivial characters, where the discriminant factors cancel in the full regulator. A proper suborbit is not treated.

(3) **Non-abelian extensions with an abelian subgroup of small index.** If $N/\mathbb{Q}$ is Galois and $\mathrm{Gal}(N/\mathbb{Q})$ has a normal subgroup with quotient C_2 whose fixed field is imaginary quadratic, the extension of that fixed field is abelian and falls under Definition 2.1. The conjugate action of $\mathrm{Gal}(k/\mathbb{Q})$ on G mixes characters. An integral statement that is equivariant for the full group $\mathrm{Gal}(N/\mathbb{Q})$, rather than for G, would require a non-abelian regulator and is open even at the level of the product formula.

(4) **Leopoldt's conjecture in the p-adic regulator.** The complex embedding λ_F was enough to compare ranks. A p-adic logarithm $E_F\otimes\mathbb{Z}_p \to X_G\otimes\mathbb{Z}_p$ would let the higher Stark module map directly to a p-adic leading element. Leopoldt's conjecture is not known for every abelian extension of an imaginary quadratic field, so no such map is used.

(5) **Auxiliary truncation sets disjoint from S_{ram}.** Rubin–Stark elements are often modified by an auxiliary set T of places, disjoint from the truncation set, in order to clear denominators at primes dividing w_F [8]. The multiplier $\prod_{v\in T}(1-\chi(\varphi_v)Nv)$ is an algebraic integer when χ is unramified at v. Inserting it into $\Theta(F/k)$ changes $\mathfrak{z}(F/k)$ by a controlled factor. The interaction of that factor with the annihilator of $\mu(F)/\mu(k)$ is not carried out here.

The three theorems are arranged so that each hypothesis is visible in the statement it supports. Theorem 1.1 has no extra hypothesis beyond the almost abelian datum. Theorem 1.2 requires $p\nmid|G|$ for the étale decomposition, and requires failure of ND_v for the vanishing that makes truncation necessary. Theorem 1.3 requires ND_v in order that the truncated index stay rational, and requires $p\nmid|G|$ for the class-group isomorphism. Removing any of these restrictions changes the shape of the result, as the Hilbert class field of $\mathbb{Q}(\sqrt{-5})$ shows for the prime dividing the group order.

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